



Course: AI and ML for Robotics

Lab Sheet 6: Sampling based and Evolutionary algorithm for Path Planning

Objective:

Implement sampling based planner and evolutionary techniques for Path Planning

Tools and Libraries Required:

Programming language: Python

Platform : Google Colab T4 GPU

Simulation : PyBullet

Design:

Input :

Obstacle list (axis-aligned rectangular obstacles with (x_min,x_max,y_min,y_max)).

Output: **Start and goal nodes** (2D coordinates).

Map size (width, height).

Roadmap (set of sampled Node objects with neighbor lists).

Planned path (ordered list of Node from start to goal found).

Visual plots showing obstacles, roadmap edges, and the final path.

Board Implementation pipeline

1. Representation of the environment and waypoints as nodes
2. Search path/waypoint in collision free region
3. Connect the neighbor points to construct the path
4. (Optionally) If the algorithm produces map, query the map using search to find the optimal path

Practice Questions:

In LabSheet 6(a):

1. Replace uniform sampling techniques used for collision check with sophisticated techniques
2. Make the connection radius more adaptive based on sample density and map size instead of fixed radius
3. Use A* with a Euclidean heuristic for faster query times when start/goal are known

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4. For dynamic replanning, integrate incremental search like D*
5. Measure success rate, planning time, path length, and clearance across many random maps and compare batch vs incremental PRM*
6. Try to impute sudden obstacles during the path building phase and observe the behaviour
7. Try other sampling based planner like RRT* and compare the performance

In LabSheet 6(b):

1. Tune the hyperparameter like generations, population size, mutation rate, cross over technique and observe the results to interpret
 2. Implement Ant colony and Particle Swarm optimization. Compare the suitability for path planning in large state/action spaces
 3. Explore Neurevolution, a way to use evolutionary algorithms to generate artificial neural networks. Can DNN learnt be used for path planning? Comment your observation.
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